

Ambiguity attitudes under matching and choice*

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Abstract

We develop a stylized noisy coding model of ambiguity attitude, under which deviations from ambiguity neutrality are determined by the cognitive frictions and sequential structure of the elicitation format rather than by stable preferences. The model predicts different and even opposite measured attitudes across formats. To test these predictions, we use a novel 10×10 symmetric stimulus design that presents identical choice problems either as standalone binary Choices or embedded in Matching Probability (MP) lists or Matching Event (ME) lists. We document two main findings. First, MP produces substantially stronger a-insensitivity (the tendency to compress subjective likelihoods toward 50–50) than standalone Choice; this stronger insensitivity is also associated with greater choice noise. Second, ME produces a-oversensitivity — the precise sign reversal of the MP pattern — despite presenting the same choice problems in an otherwise identical list structure. These findings call into question whether elicited ambiguity attitudes can be interpreted as stable preference primitives, and suggest that prominent empirical regularities can be endogenous by-products of the Bayesian encoding process underlying specific elicitation procedures.

Keywords: Noisy coding; ambiguity attitudes; a-insensitivity; matching probability/event; binary choice; procedural invariance

JEL Classification: C81; C91; D81; D91

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1 Introduction

Ellsberg (1961) famously showed that most people prefer drawing from an urn with a known composition over one whose composition is entirely unknown — a preference for known over unknown probabilities now widely documented as *ambiguity aversion*. Subsequent experimental work has established a likelihood-dependent pattern: decision makers tend to exhibit ambiguity seeking for unlikely events and ambiguity aversion for likely ones (Trautmann and van de Kuilen, 2015). This pattern is captured by the notion of ambiguity-generated likelihood insensitivity (*a-insensitivity*) — a compression of perceived likelihoods toward fifty-fifty (Fox and Tversky, 1995; Abdellaoui et al., 2011; Baillon et al., 2025). To account for these empirical regularities, a long series of models has been proposed, most of which are rooted in a utility maximization framework, forming a well-established preference-based tradition in ambiguity research.¹

Recently, a growing body of work instead attributes many canonical choice patterns to the imprecise way in which decision makers process numerical information (i.e., lottery payoffs and probabilities), rather than to stable underlying preferences (Khaw et al., 2021; Frydman and Jin, 2022; Enke and Graeber, 2023; Vieider, 2024). One implication is that elicited behavior may be sensitive to payoff-irrelevant features of the choice environment, including the procedural format in which problems are presented — a dependence that preference-based models rule out, and which therefore renders systematic format-variation a direct test of the cognitive account (Bouchouicha et al., 2025; Oprea and Vieider, 2026).

Inspired by these works, we develop a noisy coding account of how the procedural architecture of elicitation methods shapes revealed ambiguity attitudes, and derive a set of sharp, testable predictions. We first test them by using a design that presents identical choice problems either as standalone binary Choice or embedded in Matching Probability (MP) choice lists — in which subjects choose between a fixed ambigu-

¹One major family has gained particular prominence for its ability to account for likelihood-dependent ambiguity attitudes through a non-additive weighting of events or sources. Important examples include Choquet expected utility (Schmeidler, 1989), the source-dependent generalization of prospect theory (Tversky and Kahneman, 1992; Tversky and Wakker, 1995), neo-additive capacities (Chateauneuf et al., 2007), and source theory (Abdellaoui et al., 2011; Baillon et al., 2025). A second prominent family models ambiguity attitudes through a second-order transformation over utilities or expected utilities; an important example is the smooth ambiguity model (Klibanoff et al., 2005).

ous lottery and a sequence of risky alternatives with varying winning probabilities (Dimmock et al., 2016; Baillon et al., 2018). We find that both ambiguity seeking and a-insensitivity are substantially attenuated when the same problems are presented as Choice, and that a substantial share of participants reverse the sign of their measured attitude across the two methods.

More decisively, in another between-subject condition, we reverse the matching direction by swapping the risky option and the ambiguous option to derive Matching Event (ME) choice lists. As the model predicts, we find an entire reversed pattern: a-oversensitivity rather than a-insensitivity, even when the underlying problem set and the list structure (one side fixed and one side varying) are held constant. Taken together, these findings challenge the view that elicited ambiguity attitudes can be interpreted as stable, method-invariant preference primitives.

A Noisy Coding Model. Our noisy coding model shares the framework with those related models that attributes deviations from expected utility to the imprecise, Bayesian way in which decision-makers (DMs) internally represent probabilistic information (Frydman and Jin, 2023; Vieider, 2024; Bouchouicha et al., 2025; Khaw et al., 2025). Under this view, DMs combine a noisy internal signal of the true winning odds with a prior expectation in a Bayesian fashion, compressing their perception toward the prior and thereby generating systematic probability distortions endogenously. We extend this framework to the domain of ambiguity by building on Vieider (2025): because an ambiguous event is inherently harder to represent mentally than a risky one, it is encoded with greater noise and receives strictly less weight relative to the prior. This asymmetry in encoding precision thereby generates a-insensitivity endogenously without any additional preference parameters.

The procedural architecture of each elicitation method then determines how severely this compression manifests. In standalone Choice, DMs simultaneously compare the perceived likelihoods of the risky and ambiguous lotteries;² because the two lotteries are evaluated separately, the distortions in their internal representations partially offset

²For simplicity, our setup restricts attention to the case in which the two lotteries share identical outcome structures and differ only in whether the winning probability is known or unknown. Subjects therefore need only compare the winning odds of the two options, abstracting from any noise in the payoff dimension. Relaxing this restriction to allow for heterogeneous outcome structures does not qualitatively affect our theoretical predictions.

each other, yielding null to mild a-insensitivity. By contrast, MP imposes a sequential cognitive procedure: under the model, DMs first form an evaluation of the fixed ambiguous event and then search for the objective probability that matches it (see similar approaches in [Bouchouicha et al., 2025](#); [Khaw et al., 2025](#)). **This two-stage structure compounds the compression.** Since the Stage 1 anchor is already shrunk toward the prior, Stage 2 calibrates the matching probability against an already-distorted anchor — producing stronger a-insensitivity than under Choice.

The same procedural logic also predicts noisier choices under MP. In Choice, both options are evaluated simultaneously without a prior anchor, so no Stage 1 noise is introduced. The two-stage matching procedure, by contrast, feeds Stage 1 noise forward into Stage 2: DMs search for a matching probability against an anchor that is itself noisy, so decision noise accumulates across stages. In our experimental setup, this should lead to more frequent violations of basic rationality benchmarks, lower test-retest reliability, and greater between-list inconsistency.

This two-stage logic generates a further unique prediction for the ME elicitation, which reverses the roles of the two lotteries: the risky lottery is now fixed in Stage 1, and DMs search for a matching ambiguous event in Stage 2. This role reversal flips the direction of the first-stage compression, yielding a-oversensitivity rather than a-insensitivity. Beyond the sign reversal, the model delivers a subtler prediction concerning magnitude: since the risky fixed option is encoded with greater precision than an ambiguous one, the first-stage anchor is less severely compressed under ME than under MP, implying that the magnitude of a-oversensitivity under ME should fall short of that of a-insensitivity under MP. We test both predictions in a dedicated between-subject experiment.

MP versus Choice. Following a common practice, we reveal participants’ ambiguity attitudes via measuring matching probabilities in the control condition of our experiment ($N = 201$). The *matching probability* of an event is defined as the probability at which an individual is indifferent between a risky lottery that pays a prize with that probability and an ambiguous lottery that pays the same prize conditional on the event. Specifically, this baseline condition has two randomly-ordered tasks. In the MP task, matching probabilities are elicited via choice lists, each of which presents a series of

binary choices between a fixed ambiguous lottery and a risky lottery whose winning probability varies across rows, and the matching probability is inferred from the participant's response pattern across the list (e.g., [Baillon and Bleichrodt, 2015](#); [Dimmock et al., 2016](#); [Baillon et al., 2018](#)).

In the Choice task, the constituent binary choices of the MP lists are instead presented as standalone problems, one per page, in a randomized order; the matching probability is then recovered by applying the same switch-point criterion to the individually presented choices. Because both tasks draw on an identical set of underlying problems, this design provides the strongest possible test of the hypothesis that the two methods yield consistent attitude estimates. Under any standard preference-based ambiguity model, the two tasks are equivalent. Any systematic difference in elicited matching probabilities therefore points to elicitation-dependent cognitive processes, rather than to stable underlying preferences alone.

We document statistically and economically significant differences in elicited ambiguity attitudes across the two tasks. In MP, we replicate the conventional pattern of a-insensitivity widely reported in the ambiguity literature: ambiguity seeking for unlikely events and ambiguity aversion for likely ones, with elicited matching probabilities closely aligned with those in highly comparable studies (e.g., [Dimmock et al., 2015, 2016](#)). In Choice, we observe a qualitatively similar pattern, but the deviation from ambiguity neutrality is reduced by approximately 50%. Specifically, a substantial share of participants (approximately two thirds) yield inconsistent matching probability estimates across the two tasks.

Three further results speak to the relative choice noisiness of each task. First, among the 100 involved choice problems, ten have a dominant ambiguous option — namely, those in which the ambiguous lottery is degenerate, guaranteeing a prize with certainty. With the exception of the choice in which the risky option is equally degenerate, rational DMs should prefer the ambiguous option irrespective of their ambiguity attitudes. Yet participants commit significantly more violations of this dominance criterion in MP than in Choice. Second, for the repeated decision problems included as consistency checks, participants are substantially more likely to give identical answers across repetitions in Choice than in MP. Third, although MP reduces within-list multiple switching, it gives rise to substantially more between-list inconsistencies, so that

overall choices are noisier under MP than under Choice. These results paint a coherent picture: relative to Choice, MP elicits noisier and less consistent responses, with more frequent violations of basic rationality benchmarks.

ME versus MP. The two-stage noisy coding model makes a further prediction that goes beyond the MP–Choice comparison: inverting the matching direction should not merely modulate the degree of a-insensitivity but reverse its sign entirely. To test this prospectively, we introduce the ME task as a between-subject complement to MP: a separate sample of 199 participants is assigned to ME rather than MP, with all other design elements — particularly the same 100 underlying decision problems — held constant. The results confirm both the sign-reversal and the magnitude-asymmetry predictions. Participants exhibit a-oversensitivity — ambiguity aversion for unlikely events and ambiguity seeking for likely ones — the precise mirror image of the a-insensitivity pattern documented under MP. And as the model further predicts from the greater encoding precision of risky options, the magnitude of this deviation is significantly smaller under ME than under MP. These results provide prospective validation of the two-stage mechanism, confirming a prediction that no existing preference-based theory of ambiguity has anticipated.

Implications. These findings carry three implications that we develop in the discussion. First, the ME sign reversal poses a direct challenge to preference-based accounts: since MP and ME present the same underlying problems in an identical list format (one side fixed and another side varying), no rescaling of preference parameters can accommodate a sign reversal in measured attitudes — suggesting that a-insensitivity in MP and a-oversensitivity in ME reflect the cognitive architecture of the elicitation format rather than a stable preference primitive. Second, by endogenizing the weighting objects of popular descriptive frameworks — such as the capacity function and the source function — as outputs of a Bayesian encoding process, the noisy coding model points toward a productive complementarity between cognitive accounts and descriptive ambiguity models. Third, if ambiguity attitudes are shaped by cognitive limitations rather than stable tastes, then correlations between ambiguity attitudes and field outcomes should be interpreted with caution, as they may reflect cognitive ability and elicitation-induced distortion as much as variation in underlying preferences.

2 Related Literature

This paper intersects with various bodies of literature. It first belongs to the emerging literature on noisy cognition in economic decision-making, which attributes systematic choice patterns to imprecise internal representations rather than stable preferences. It also contributes to the procedural invariance literature by documenting fresh format-dependent reversals in the ambiguity domain with offering unique experimental design and novel theoretical explanations.

Noisy Cognition in Decision-Making. A growing literature attributes well-documented patterns in decision-making under risk — including the fourfold pattern of risk attitudes and the inverse-S shape of probability weighting — to the imprecise, Bayesian way in which decision makers internally represent lottery information (Natenzon, 2019; Khaw et al., 2021; Enke and Graeber, 2023; Frydman and Jin, 2023; Vieider, 2024; Netzer et al., 2025). Under this view, agents combine a noisy internal signal of the true stimulus with a prior expectation, compressing their perception toward the prior and generating systematic distortions endogenously — without requiring those distortions to be stable features of the preference function. Our paper contributes to this literature by extending the noisy coding framework to the domain of ambiguity. Because an ambiguous event is inherently harder to represent mentally than a risky one, it is encoded with greater noise and receives less weight relative to the prior. This asymmetry generates a-insensitivity endogenously, and — as we show — produces testable predictions about how the direction and magnitude of the distortion depend on the procedural architecture of the elicitation task.

The paper most directly related to ours is Bouchouicha et al. (2025), who study format effects on risk attitudes using the same methodological contrast between a pricing list task and the constituent binary choices extracted from it. They find that the fourfold pattern of risk attitudes is prominent under the list format but largely collapses under binary choice, and attribute this divergence to the sequential evaluation structure of the list, which amplifies cognitive distortions relative to the simultaneous comparison in binary choice — a mechanism we adopt and formalize.

Our paper makes two theoretical advances beyond that framework. First, the en-

coding precision asymmetry between risky and ambiguous events generates a prediction with no counterpart in the pure-risk setting: because the ambiguous event is the harder-to-encode option, the two-stage structure of the matching task compounds the compression more severely, predicting that a-insensitivity will be substantially more pronounced under MP than under Choice. Second, because our model concerns the comparison between two likelihoods — one known, one unknown — it naturally accommodates both MP and ME as special matching cases. The two scenarios differ only in which lottery is fixed, and the model generates sharply distinct predictions for each: the compression asymmetry that produces a-insensitivity under MP reverses direction entirely under ME, yielding a-oversensitivity. This sign reversal prediction, together with the further prediction that its magnitude under ME falls short of that under MP, follows directly from the model mechanism and constitutes a novel contribution to the growing literature on the cognitive foundations of economic decision-making under uncertainty.

Procedural Invariance in Preference Elicitation. A substantial literature tests procedural invariance in decision-making under risk, with a recurring finding that measured risk attitudes are sensitive to the elicitation method employed (e.g., [Lichtenstein and Slovic, 1971](#); [Harbaugh et al., 2010](#); [Loomes and Pogrebna, 2014](#); [Pedroni et al., 2017](#)). A particularly related strand of this work compares probability equivalents (PE) — where the certain amount and lottery payoffs amounts are fixed and subjects search for the equivalent lottery probability — against certainty equivalents (CE) — where the lottery is fixed and subjects search for the equivalent sure amount. [Hershey and Schoemaker \(1985\)](#) first established that these two methods yield systematically different risk attitude estimates. [Sprenger \(2015\)](#) revived this comparison within a model of stochastic reference dependence, showing that subjects exhibit more risk aversion under PE than CE tasks. More recently, [Feldman and Ferraro \(2025\)](#) document that preference reversals between CE and PE are probability-dependent: subjects are more risk-tolerant under CE for unfavorable odds, but this direction reverses near certainty, where CE instead induces greater risk aversion. [Shubatt and Yang \(2026\)](#) rationalize this reversal through a comparison complexity framework, in which systematic pull-to-center distortions arise from the difficulty of aggregating tradeoffs between outcome and probability dimensions across the two valuation formats.

Our findings are superficially parallel — MP elicitation generates a-insensitivity and ME reverses the sign — but the mechanism differs fundamentally. In the CE/PE literature, both options involve known probabilities, and distortions arise from comparing a lottery against an incongruent sure-amount price list. In our setup, both options share identical prize structures and differ only in whether the winning event is risky or ambiguous; there is no outcome-probability tradeoff. The sign reversal and the magnitude asymmetry between MP and ME we document therefore cannot be attributed to tradeoff complexity (Shubatt and Yang, 2026) or reference dependence (Sprenger, 2015; Feldman and Ferraro, 2025) over a fixed sure amount, but follows instead from the differential encoding precision of risky and ambiguous events — an ambiguity-specific source of format-dependent distortion that is logically independent of the mechanisms proposed for the risk domain.

Another strand talks about preference reversals between choice and valuation, and on this front, Maafi (2011) and Trautmann et al. (2011) are the only studies we are aware of. Studying the p -bet and \$-bet setup, Maafi finds that most participants choose the p -bet in binary choice yet over 85% assign a higher price to the \$-bet in a valuation task — a reversal attributable to strong ambiguity seeking for small-likelihood events under the valuation format, consistent with the distortions our model predicts. Trautmann et al. (2011) document a similar reversal: roughly 30% of participants choose the ambiguous lottery in binary choice, yet about half of that group assign it a lower willingness to pay. Both studies, however, are limited to a single moderate likelihood and do not employ an identical problem set across methods, precluding a characterization across the full probability range or cardinal comparisons of magnitude. Our design instead are free from both limitations, and we show that the direction and magnitude of preference reversals depend critically on the likelihood level — a pattern neither prior study could detect or explain.³

³Trautmann et al. (2011) propose a reference-dependence account in which participants anchor on the risky bet's value and adjust downward for the ambiguous one due to loss aversion. This cannot explain ambiguity seeking for small likelihoods under MP, nor the reversed pattern under ME.

3 Choices under Ambiguity as Bayesian inference

Standard preference-based models treat ambiguity attitudes as stable primitives. Consequently, elicited attitudes should be invariant to the format in which a decision problem is presented, provided that the underlying choice objects are held fixed. Recent evidence from the risk domain challenges this invariance prediction. [Bouchouicha et al. \(2025\)](#) show that the fourfold pattern of risk attitudes is pronounced when choices are embedded in a list format, but largely disappears when the same constituent decisions are presented as standalone binary choices. They attribute this divergence to the sequential evaluation structure of list-based elicitation, which amplifies cognitive distortions relative to simultaneous binary comparison.

Building on this mechanism, we extend the noisy-coding framework to decision-making under ambiguity. The central idea is that observed ambiguity attitudes need not directly reveal stable preferences; instead, they may arise from Bayesian inference over imperfect internal representations of risky and ambiguous likelihoods ([Vieider, 2024, 2025](#)). This section develops the modelling framework to derive predictions across different elicitation procedures. To keep the exposition focused, the main text emphasizes the economic intuition and testable implications of the model, while the full technical derivations are deferred to [Appendix A](#).

3.1 Belief Evaluation

The central premise of noisy coding is that DMs cannot directly access probabilities or event likelihoods with perfect precision. Instead, they form noisy internal representations of these quantities and combine them with prior expectations in a Bayesian manner. Following [Vieider \(2024, 2025\)](#), we assume that before entering the choice process, the DM mentally represents each winning probability p through a noisy signal r_p :

$$r_p \sim \mathcal{N}(\ell(p), \nu_p^2)$$

where $\ell(p) \equiv \log \frac{p}{1-p}$ denotes the log-odds transformation.⁴ The parameter ν_p captures coding noise and therefore the degree of imprecision in the internal representation of probabilities. Larger values of ν_p correspond to noisier signals and less precise perception. Such imprecision arises from the limited cognitive resources available for information processing, particularly when assessments are formed quickly and intuitively rather than derived from an optimal computational procedure (Woodford, 2020). The noisy signal is then integrated with a prior belief over log-odds, given by $\mathcal{N}(\mu, \sigma^2)$, to form the actionable posterior belief that enters decision stage. Bayesian inference implies that the posterior expectation is a precision-weighted average of the signal and the prior mean:

$$\mathbb{E}[\ell(p) \mid r_p] = \alpha_p r_p + (1 - \alpha_p) \mu$$

where $\alpha \triangleq \frac{\sigma^2}{\sigma^2 + \nu_p^2}$ captures the weight placed on the perceived signal relative to the prior mean and is determined by the precision of the signal relative to the prior. When the internal representation is highly precise, ν_p^2 is small and α approaches one, so posterior beliefs rely primarily on the signal. Conversely, when the signal is noisy, α is smaller and posterior beliefs are pulled more strongly toward the prior mean.

Because ambiguous events are inferred with greater uncertainty than risky ones, the elicitation procedure determines how this asymmetry manifests in observed choices. In what follows, we show that this framework generates sharp predictions across elicitation methods.

3.2 Standalone Binary Choice

We begin with the benchmark case of a standalone binary choice between a risky lottery and an ambiguous lottery. The risky lottery pays prize x with objective winning probability p_r , whereas the ambiguous lottery pays the same prize conditional on an ambiguous event E_a , whose ambiguity-neutral probability is p_a . Thus, the two lotteries differ only in the source of uncertainty: the winning probability is known for the risky

⁴Here, p refers to the objective probability for the risky option and the a-neutral probability for the ambiguous option. We adopt the logit-normal specification primarily for expositional convenience, as its conjugacy with the normal signal structure yields closed-form posterior expressions. Mapping probability distributions into log-odds space is also consistent with the neural coding framework, in which choice-relevant quantities are represented in log-odds units (Green and Swets, 1966; Gold and Shadlen, 2001, 2002). Our qualitative conclusions, however, do not depend on this particular distributional assumption and extend to a broad class of single-peaked distributions.

lottery but unknown for the ambiguous lottery.

The key assumption is that ambiguous likelihoods are encoded with greater imprecision than risky probabilities. Formally, we assume $\nu_a > \nu_r$, where ν_a and ν_r denote the coding noise associated with the ambiguous and risky options, respectively. This captures the idea that inferring an unknown event likelihood is cognitively more demanding than representing an objective probability. Consequently, the DM assigns lower confidence to the inferred likelihood for the ambiguous urn than to the objective probability associated with the risky urn (Vieider, 2025).

In a standalone binary choice, the DM evaluates the two options independently and simultaneously. Given the identical payoff scheme, one chooses the risky lottery whenever its posterior expected log-odds exceed those of the ambiguous lottery:

$$\mathbb{E}[\ell(p_r) \mid r_r] > \mathbb{E}[\ell(p_a) \mid r_a]$$

This yields the following stochastic choice rule:

$$\Pr[(x, p_r; 0) \succ (x, E_a; 0)] = \Phi \left(\frac{\rho \ell(p_r) - \gamma \ell(p_a) + (\gamma - \rho)\mu}{\sqrt{\rho^2 \nu_r^2 + \gamma^2 \nu_a^2}} \right) \quad (1)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The posterior weights are given by $\rho \triangleq \frac{\sigma^2}{\sigma^2 + \nu_r^2}$ and $\gamma \triangleq \frac{\sigma^2}{\sigma^2 + \nu_a^2}$. Since the ambiguous signal is encoded with lower precision than the risky signal, it follows that $0 < \gamma < \rho < 1$. Thus, the ambiguous signal is processed with less confidence than the risky signal and is correspondingly shrunk more strongly toward the prior.⁵ For each ambiguous event E_a with ambiguity-neutral probability p_a , its matching probability $\bar{p}_r$ under the binary choice format can be characterized by the stochastic indifference point.

Prediction 1 (Choice a-insensitivity.) *In standalone Choice, the matching probability is compressed toward the prior:*

$$\ell(\bar{p}_r) = \mu + \frac{\gamma}{\rho} (\ell(p_a) - \mu)$$

Since $0 < \gamma/\rho < 1$, the matching probability inferred from Choice is a compressed transformation of the ambiguity-neutral probability: low ambiguous probabilities are shifted upward

⁵Here we assume a common prior for risky and ambiguous likelihoods, reflecting the symmetry of the experimental design. This assumption is not essential for the predictions derived throughout the paper. If priors are allowed to differ across risk and ambiguity, the same qualitative predictions obtain whenever the ambiguous signal is less precise relative to its prior variance, that is, whenever $\nu_a/\sigma_a > \nu_r/\sigma_r$.

toward the prior, whereas high ambiguous probabilities are shifted downward.

The model therefore generates a-insensitivity endogenously, without introducing ambiguity attitudes as separate preference primitives. Notably, after rearrangement, it can be expressed in forms of:

$$\frac{\ell(\bar{p}_r) - \mu}{\ell(p_a) - \mu} = \frac{\gamma}{\rho} \quad (2)$$

which further showcases the relative intensity of shrinkage between these two likelihoods.

3.3 Choice List Elicitation

We now extend the baseline model to choice-list environments by following the two-stage cognitive procedure proposed by [Bouchouicha et al. \(2025\)](#) for the case of certainty equivalent choice lists.⁶ In both the MP and ME formats, the list structure naturally separates the task into two components. The DM first evaluates the option that remains fixed throughout the list, thereby forming an internal anchor. She then searches across the varying rows for the option whose perceived signal most closely matches this anchor.

3.3.1 MP Choice List

In an MP list, the DM faces a sequence of risky lotteries with objective winning probabilities $p_{r_1} < p_{r_2} < \dots < p_{r_j}$, each of which is benchmarked against the same ambiguous lottery. The ambiguous lottery is fixed throughout the list and is associated with an ambiguity-neutral probability p_a . Since the ambiguous option is prominently displayed and remains constant across rows, it is natural to assume that the DM first evaluates this fixed option and then searches for the risky probability that best matches its perception.

Stage 1: Evaluating the fixed option. In the first stage, the DM forms an initial evaluation of the fixed ambiguous lottery. As in the standalone Choice case, this evalua-

⁶Because the present setting involves comparisons between probabilities rather than between monetary outcomes and probabilities, the analysis remains entirely on the log-odds scale. This makes the extension structurally more straightforward than the certainty-equivalent case in [Bouchouicha et al. \(2025\)](#), while preserving the central insight that list elicitation can change how noisy internal representations are translated into observed responses.

tion is centered on the posterior perception of its log-odds, given by $\mathbb{E}[\ell(p_a) \mid r_a] = \gamma\ell(p_a) + (1 - \gamma)\mu$, which then serves as the internal anchor for the subsequent search across the varying risky probabilities.

Stage 2: Searching across the varying rows. In the second stage, the DM searches across the risky probabilities in the list for the row whose internal signal most closely matches the ambiguous anchor inferred from Stage 1. This procedure differs fundamentally from the standalone binary-choice environment. Under standalone Choice format, the risky and ambiguous options are evaluated simultaneously and independently on the basis of their respective signals, and their posterior evaluations are then compared directly. Under the MP format, by contrast, the fixed ambiguous option is evaluated first and transformed into an internal anchor. The DM subsequently searches across the risky probabilities in the list for the signal that produces indifference with that anchor. The elicited matching probability is therefore generated by a sequential anchoring-and-search procedure rather than by a direct bilateral comparison of independently evaluated options. Under a continuous approximation, this search problem is represented by

$$p_r^* = \arg \min_{p_r} \|r_r \mid p_r - [\gamma\ell(p_a) + (1 - \gamma)\mu]\|$$

This search problem selects the risky probability p_r^* whose log-odds match the ambiguous anchor on average, thereby yielding the following predictions.

Prediction 2a (MP a-insensitivity.) *In an MP list, the latent matching probability satisfies*

$$\frac{\ell(p_r^*) - \mu}{\ell(p_a) - \mu} = \gamma \quad (3)$$

Since $0 < \gamma < 1$, the MP response also exhibits a-insensitivity.

Prediction 2b (Stronger compression under MP than under Choice.) *Since $0 < \rho < 1$, we have $\gamma < \gamma/\rho$. Therefore, relative to standalone binary Choice, MP lists amplifies a-insensitivity.*

3.3.2 ME Choice List

In an ME list, the matching direction is reversed. The fixed option is now a risky lottery with objective winning probability p_r , whereas the varying options are ambiguous

lotteries associated with ambiguity-neutral probabilities $p_{a_1} < p_{a_2} < \dots < p_{a_J}$.

As in the MP list, the fixed option is evaluated first and then serves as the internal anchor for the subsequent search across the varying rows. The crucial difference is that this anchor is now risky rather than ambiguous. The DM therefore selects the ambiguous event whose ambiguity-neutral probability p_a^* matches this risky anchor best, yielding the following predictions.

Prediction 3a (ME a-oversensitivity.) *In an ME list, the latent matching event satisfies*

$$\frac{\ell(p_r) - \mu}{\ell(p_a^*) - \mu} = \frac{1}{\rho} \quad (4)$$

Since $1/\rho > 1$, the ME response exhibits a-oversensitivity rather than a-insensitivity.

The intuition is that reversing the matching direction also reverses the list-induced distortion. As a result, the inferred ambiguous event required to match a given risky probability must adjust in the opposite direction. The model therefore predicts a pattern of a-oversensitivity in ME lists: ambiguity aversion for unlikely events and ambiguity seeking for likely events.

Prediction 3b (Stronger distortion under MP than under ME.) *The distortion induced by MP lists is stronger in magnitude than the reversed distortion induced by ME lists.*

The insensitivity generated by the MP list is governed by $1 - \gamma$, whereas the oversensitivity generated by the ME list is governed by $1 - \rho$. Since ambiguous likelihoods are encoded with lower confidence than risky probabilities, we have $0 < \gamma < \rho < 1$. It follows that $1 - \gamma > 1 - \rho$, thus the prior-centered compression induced by MP lists is stronger than the reversed distortion induced by ME lists. This asymmetry arises because MP anchors on the fixed ambiguous option, whose signal is more heavily shrunk toward the prior, whereas ME anchors on the fixed risky option, whose signal is comparatively more precise and therefore less strongly shrunk.

3.3.3 Decision Noise

Beyond accounting for distortions in measured ambiguity attitudes across elicitation formats, the noisy-coding model also yields novel predictions regarding decision noise. According to the stochastic choice rule presented in Equation (1), the variance term in

the denominator captures decision noise in standalone binary choice as

$$V_{BC} = \rho^2 \nu_r^2 + \gamma^2 \nu_a^2$$

Under the same structural normalization, the corresponding decision noises under the list-based procedures are

$$V_{MP} = \nu_r^2 + \gamma^2 \nu_a^2, \quad V_{ME} = \nu_a^2 + \rho^2 \nu_r^2$$

for MP and ME, respectively; see Appendix A for the detailed derivations. Since $\rho < 1$ and $\gamma < 1$, both MP and ME generate strictly greater decision noise than standalone binary choice.

Prediction 4 (Greater decision noise under choice lists.) *Both MP and ME induce noisier responses than standalone binary choice.*

It implies that the stronger distortions observed under list-based elicitation need not reflect a more accurate revelation of underlying ambiguity preferences. Rather, the increase in decision noise is consistent with the propagation of errors through the two-stage decision strategy induced by the list structure.

4 Elicitation-Dependent Ambiguity Attitudes

In this section, we experimentally examine whether decision makers exhibit systematic differences in ambiguity attitudes across three elicitation methods: standalone binary choice (BC), matching probability (MP), and matching event (ME). Our analysis provides new and comprehensive evidence that ambiguity preferences reverse across distinct presentational contexts, in line with the predictions of the Bayesian inference framework developed above. We further investigate choice suboptimality and inconsistency, which provide a powerful perspective for distinguishing our noisy-coding model from existing models of ambiguity (e.g., [Klibanoff et al., 2005](#); [Baillon et al., 2025](#)).

4.1 Experimental Design

Basic Standalone Choices. The core of the experiment employs a classic two-urn design inspired by [Ellsberg \(1961\)](#), which constitutes the most widely used paradigm

Please choose one box to draw a ball



Figure 1: An example of a binary choice problem. The left (risky) box yields an objective winning probability of $p = 0.5$. The right (ambiguous) box yields an unspecified winning probability that could be anywhere in $[0, 1]$. Both options pay \$20 for a winning draw and nothing otherwise.

for eliciting ambiguity attitudes in a controlled laboratory setting. Each decision problem presents participants with two boxes, each containing exactly ten balls numbered 0 through 9. Participants are asked to choose one box from which a ball will be drawn at random, with a constant monetary payoff of \$20 determined by whether the drawn ball falls within a pre-specified winning range. Figure 1 provides an example of such decision problems.

The two boxes differ in one critical dimension: the composition of balls in the first box is fully transparent to the participant, whereas the composition of balls in the second box is concealed. Specifically, the *risky box* is visually transparent, displaying all ten balls with their numbers explicitly visible. This design ensures that the winning probability associated with any given numerical range is objectively and unambiguously determined. The *ambiguous box*, by contrast, is opaque: while participants are informed that it similarly contains ten balls, the exact numerical composition is unknown, rendering the winning probability subjectively unspecified. The winning probability for the ambiguous box could, in principle, be anywhere in the interval $[0, 1]$ —it may be higher than, equal to, or lower than the winning probability of the risky box.

In this example given by Figure 1, both boxes are associated with the same winning condition — drawing a ball numbered 0 through 4 — which corresponds to an objective winning probability of $p = 0.5$ for the risky box. Both options offer an identical prize upon winning and nothing otherwise. Because the two options are monetarily equivalent in terms of prize structure and share the same stated winning range, any systematic preference between them can be attributed solely to the participant's attitude

toward ambiguity. A participant who chooses the risky box in this situation reveals a preference for known over unknown probabilities, and is thus classified as (weakly) ambiguity averse. Conversely, a participant who chooses the ambiguous box reveals a preference for unknown over known probabilities, and is classified as (weakly) ambiguity seeking.

This setup provides a clean and theoretically grounded framework for identifying heterogeneity in ambiguity attitudes at the individual level. The binary choice format minimizes strategic complexity and cognitive demands, making the task accessible while preserving its power to discriminate among distinct behavioral patterns. Crucially, holding the prize structure constant across the two options ensures that observed preferences reflect probability evaluation rather than outcome valuation, which is essential for a clean identification of ambiguity attitudes.

Matching Probability Choice Lists. The basic binary choice described above reveals only an ordinal preference between the risky and the ambiguous box. To obtain a cardinal measure of ambiguity attitude, the individual subjective probability of ambiguous events needs to be measured. The most widely adopted elicitation method is the matching probability (MP) (Kahn and Sarin, 1988). Specifically, the matching probability of an ambiguous event E_a (with a-neutral probability as p_a) is defined as the objective winning probability $m(p_a) \in [0, 1]$ at which a participant is indifferent between betting on E_a and betting on the risky box with winning probability $m(p_a)$. Formally, for a fixed prize of \$20, the matching probability $m(p_a)$ satisfies $(\$20, E_a; 0) \sim (\$20, m(p_a); 0)$. Under ambiguity neutrality, $m(p_a)$ coincides with the objective probability of the symmetric risky counterpart. Systematic deviations of $m(p_a)$ below (above) this benchmark indicate ambiguity aversion (seeking).

Following established practice (e.g., Dimmock et al., 2016; Baillon et al., 2018), matching probabilities in the current study are elicited via choice lists, as illustrated in Figure 2. Each list presents a fixed ambiguous box on the left-hand side, paired with a series of risky boxes on the right-hand side whose winning range varies across rows. The list contains ten rows in total, with the winning range of the risky box expanding monotonically from a single ball (0, corresponding to $p_r = 0.1$) at the top to all ten balls (0–9, corresponding to $p_r = 1.0$) at the bottom. In each row, participants make a

Please choose one box to draw a ball

	The unknown box	The known box	
 If the ball is 0-4, you win \$20	☐	☐	If the ball is 0, you win \$20
	☐	☐	If the ball is 0-1, you win \$20
	☐	☐	If the ball is 0-2, you win \$20
	☐	☐	If the ball is 0-3, you win \$20
	☐	☐	If the ball is 0-4, you win \$20
	☐	☐	If the ball is 0-5, you win \$20
	☐	☐	If the ball is 0-6, you win \$20
	☐	☐	If the ball is 0-7, you win \$20
	☐	☐	If the ball is 0-8, you win \$20
	☐	☐	If the ball is 0-9, you win \$20



Figure 2: Screenshot of the ascending matching probability choice list interface. Each row presents a binary choice between the ambiguous box (left, fixed across rows) and a risky box (right) whose winning range expands monotonically from a single ball (0) at the top to all ten balls (0-9) at the bottom. The corresponding objective winning probabilities range from $p = 0.1$ to $p = 1.0$ in increments of 0.1.

binary choice between the ambiguous option and the corresponding risky option. As the winning range expands, the risky option becomes progressively more attractive, so a participant is generally expected to switch from preferring the ambiguous box to preferring the risky box somewhere between two consecutive rows as she moves down the list. To mitigate any potential order effects arising from the list format, the direction of monotonicity is varied between participants: some participants face *ascending* lists in which the winning range of the risky box expands from top to bottom, while others face *descending* lists in which it contracts.

To facilitate comparison with the standalone binary-choice task introduced later, participants are required to respond to every row before a list can be submitted — that is, each list requires at least ten clicks to complete. Prior to submission, participants may freely revise any of their choices within the list; once submitted, however, no further changes are permitted. Because our design accommodates multiple preference switches, we use the proportion of rows in which the participant selects the ambiguous box as a continuous approximation of the underlying switching point, and take the corresponding objective probability as the estimated matching probability. As [Baillon et al. \(2018\)](#) argue, the choice list format encourages considered and internally consis-

tent judgments while imposing minimal cognitive burden relative to direct probability elicitation methods.

Matching Event Choice Lists. The MP approach holds the ambiguous event fixed and varies the objective winning probability of the risky option until indifference is reached. A natural and symmetric counterpart is the matching event (ME), which reverses this logic: the risky option is now held fixed at a given objective winning probability p_r , and the winning range of the ambiguous option is varied across rows until the participant is indifferent. Formally, for a fixed risky winning probability p_r , the matching event $E_{m'(p_r)}$ satisfies $(\$20, p_r; 0) \sim (\$20, E_{m'(p_r)}; 0)$, where $m'(p_r)$ is the a-neutral probability of the ambiguous box at which the participant is indifferent between the two options. Under ambiguity neutrality, $m'(p_r)$ corresponds to the numerically symmetric range on the ambiguous box, that is, the range whose cardinality implies the same winning probability as p_r . For example, given $p_r = 0.5$, the ambiguity-neutral matching event corresponds to the winning range 0–4. Systematic deviations of $m'(p_r)$ above (below) this symmetric benchmark indicate that the participant demands a larger (smaller) ambiguous winning range to accept the ambiguous option, consistent with ambiguity aversion (seeking).

While MP and ME are conceptually dual to each other — MP maps an ambiguous event to an equivalent objective probability, whereas ME maps an objective probability to an equivalent ambiguous event — their behavioral and cognitive implications need not be identical. In particular, the two formats differ in what is held fixed and what is varied during deliberation, which can engage distinct cognitive processes and yield systematically different preference expressions (Tversky and Kahneman, 1981). Examining both elicitation formats thus allows us to assess the robustness of ambiguity attitude measurements to the direction of preference elicitation, which is one of the central methodological contributions of this paper.

Figure 3 illustrates the descending version of the ME choice list. Each row presents a binary choice between the fixed risky box on the left — whose winning condition is 0–4, corresponding to $p_r = 0.5$ — and the ambiguous box on the right, whose winning range contracts monotonically from 0–9 at the top to 0 at the bottom. As the winning range of the ambiguous option shrinks, it becomes progressively less attractive, so a

Please choose one box to draw a ball

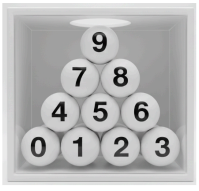
	The known box	The unknown box	
 If the ball is 0–4, you win \$20	☐	☐	If the ball is 0–9, you win \$20
	☐	☐	If the ball is 0–8, you win \$20
	☐	☐	If the ball is 0–7, you win \$20
	☐	☐	If the ball is 0–6, you win \$20
	☐	☐	If the ball is 0–5, you win \$20
	☐	☐	If the ball is 0–4, you win \$20
	☐	☐	If the ball is 0–3, you win \$20
	☐	☐	If the ball is 0–2, you win \$20
	☐	☐	If the ball is 0–1, you win \$20
	☐	☐	If the ball is 0, you win \$20

Figure 3: Screenshot of the descending matching event choice list interface. Each row presents a binary choice between the risky box (left, fixed across rows) and an ambiguous box (right) whose winning range contracts monotonically from all ten balls (0–9) at the top to a single ball (0) at the bottom. The corresponding ambiguity-neutral symmetric winning probabilities range from $p = 1$ to $p = 0.1$ in increments of 0.1.

participant is generally expected to switch from preferring the ambiguous box to preferring the risky box at some point as she moves down the list. All remaining logistics follow those of the MP lists described above.

Stimuli. The full stimulus set is defined by the Cartesian product of the two boxes' winning ranges. Each box admits ten possible winning ranges — 0, 0–1, 0–2, ..., 0–9 — yielding a total of $10 \times 10 = 100$ distinct binary choice problems. The MP and ME formats differ not in the underlying choices presented, but in how these 100 problems are organized into lists. Under MP, choices are partitioned into ten lists by fixing the ambiguous winning range and varying the risky winning range across rows. Under ME, the same 100 choices are partitioned along the orthogonal dimension: the risky winning range is fixed and the ambiguous winning range varies. Consequently, participants in both conditions encounter an identical set of 100 binary choices; the two formats differ solely in how those choices are grouped and presented, ensuring that any difference in elicited ambiguity attitudes reflects the effect of the elicitation format itself rather than differences in the underlying stimulus set.

Sample and Incentives. The experiment was conducted on Prolific, using a demographically representative sample of the United States adult population. A total of 400 participants were retained after excluding 9 whose experiment duration fell below 800 seconds, a threshold used to screen out inattentive responses. The median completion time was approximately 30 minutes, with a median age of 47 and 198 female participants. Participants received a fixed participation fee of \$6. Monetary incentives for the decision tasks were implemented via a between-subject random incentive system: participants were informed that one in every ten participants would be randomly selected to have one of their choices played out for real, with the selected participant winning \$20.⁷

Procedure. After reading the experiment instructions and completing a set of training questions, participants were required to pass three comprehension check questions before proceeding to the choice tasks. To encourage effort, an additional \$0.15 was awarded for each question answered correctly on the first attempt.⁸ Participants were then randomly assigned to either the MP condition ($n = 201$) or the ME condition ($n = 199$). Each participant completed two choice tasks: the *matching* task, in which choices were organized into lists as described above, and the *binary choice* task, in which the same 100 binary choice problems were presented *individually* on separate pages. The order of the two tasks was randomized and balanced across conditions. Between the two tasks, participants completed 15 cognitive ability test questions, serving as a break to limit fatigue. To enable consistency checks, the list corresponding to the fixed option with winning range 0–1 was presented twice within the matching task, and the corresponding ten binary choices were likewise repeated in the binary choice task. Upon completing both choice tasks, participants filled out a brief demographic questionnaire to conclude the experiment.

⁷Although Li and Vieider (2026) document that hypothetical task produce no pronounced differences relative to real incentives in decisions under risk and over time, comparable evidence is lacking for the ambiguity domain. Therefore, we continue to implement real incentives scheme in the present study. This between-subject variant of the random incentive system is a well-established practice in large-scale online experiments, preserving incentive compatibility while keeping implementation costs tractable (e.g., Abdellaoui et al., 2011; Baillon et al., 2018). Importantly, participants were informed at the very beginning of the experiment that both their bonus eligibility and the ambiguous box’s composition had been predetermined by the computer. Beyond enhancing trust in the procedure, this approach theoretically prevents strategic hedging across choices (Baillon et al., 2022).

⁸86% of participants passed all three comprehension questions correctly on their first attempt. Our findings remain essentially unchanged after excluding participants who failed at least one of the three comprehension questions (see results in Appendix C).

4.2 Binary Choice versus Matching Probability

A central feature of our design is that participants face the same underlying stimulus set in both the MP and Choice tasks. From a payoff perspective, the two tasks therefore implement the same binary comparisons; what differs is only the elicitation format. This constitutes a particularly stringent test of procedural invariance for preference-based theories of ambiguity, which would generally predict consistent behavioral patterns across payoff-equivalent tasks (e.g., [Klibanoff et al., 2005](#); [Baillon et al., 2025](#)). The richness of the stimulus set further allows us to examine elicitation effects over the full likelihood range, from unlikely to likely events. We proceed in two steps. First, we compare ambiguity attitudes across MP and Choice, focusing on whether the two formats generate different degrees of ambiguity-insensitivity as Bayesian inference model predicts. Second, we examine choice stochasticity and internal consistency to assess whether the stronger pattern observed under list elicitation reflects cleaner preference revelation or additional procedural noise.

Ambiguity Attitude. Following conventional practice ([Kahn and Sarin, 1988](#); [Dimmock et al., 2016](#)), we first derive *event-specific* ambiguity attitude (AA) indexes. Each index AA_{p_a} captures the level of “local” ambiguity aversion for an ambiguous event whose objective winning probability p_a would be used by an ambiguity-neutral decision maker — hereafter referred to as the *ambiguity-neutral* (or *a-neutral for short*) *likelihood*. Formally,

$$AA_{p_a} = p_a - m(p_a)$$

where $m(p_a)$ denotes the matching probability elicited for the ambiguous event with a-neutral likelihood p_a .⁹ A positive (negative) value of AA_{p_a} indicates ambiguity aversion (seeking) at likelihood p_a , with the elicited matching probability falling below (above) the a-neutral benchmark. The a-insensitivity pattern corresponds to a configuration in which AA_{p_a} is negative for small likelihoods — reflecting ambiguity seeking — and positive for moderate to large likelihoods — reflecting ambiguity aversion — tracing out an inverse-S shaped profile across the probability range.

⁹We use the midpoint of the switching range, approximated by the choice proportion, to derive matching probabilities. Specifically, for an ambiguous lottery ($\$20, E; 0$), if a participant selects the ambiguous option in 3 out of the 10 corresponding rows, her matching probability for event E is recorded as 0.35.

To complement the event-specific analysis, we also compute two global indexes following [Dimmock et al. \(2016\)](#). For each participant, a best-fitting line $m(p_a) \mapsto c + sp_a$ is estimated between the matching probabilities $m(p_a)$ and the a-neutral likelihoods p_a across the full probability range. The global index of a-insensitivity is defined as $a = 1 - s$, capturing the flatness of the matching probability profile — a tendency to compress all likelihoods toward 50-50. The global index of ambiguity aversion is defined as $b = 1 - s - 2c$, capturing the average elevation of the profile relative to the ambiguity-neutral benchmark. By construction, a and b are orthogonal, summarizing the curvature and level of the matching function respectively, and a positive value of each index indicates a-insensitivity and ambiguity aversion.

Figure 4 plots the event-specific ambiguity attitude indexes AA_{p_a} across all nine a-neutral likelihoods for both elicitation formats. It provides clear evidence of the classical ambiguity-insensitivity pattern under both binary Choice and MP, while also showing how this pattern varies systematically with the elicitation method, in line with the model predictions. Under the MP format, AA_{p_a} is substantially negative for low likelihoods — reaching approximately -0.14 at $p_a = 0.1$ — crosses zero near $p_a = 0.4$, and rises to roughly 0.22 at $p_a = 0.9$, closely replicating the event-specific indexes reported by [Dimmock et al. \(2016\)](#). This systematic sign reversal — ambiguity seeking for unlikely events and ambiguity aversion for likely ones — is the hallmark of a-insensitivity: decision makers fail to discriminate sufficiently among likelihood levels, compressing their subjective likelihood judgments toward fifty-fifty ([Dimmock et al., 2016](#); [Baillon et al., 2018](#)). This pattern is substantially attenuated under the Choice format, the AA_{p_a} indexes lie systematically closer to zero across virtually the entire likelihood range. Both tails are attenuated by roughly 50% — weaker ambiguity seeking at low likelihoods ($AA_{0.1}^{BC} = -0.06$ vs. $AA_{0.1}^{MP} = -0.14$) and weaker ambiguity aversion at high likelihoods ($AA_{0.9}^{BC} = 0.12$ vs. $AA_{0.9}^{MP} = 0.22$) — yielding a flatter overall profile. This symmetric compression is precisely the signature of reduced a-insensitivity rather than a uniform shift in the level of ambiguity aversion per se.

Beyond estimating the aggregate effect pooled within each elicitation method, the within-subject design enables us to examine whether the preference AA_{p_a} remains stable across methods at the individual level. For events with small a-neutral probability — $p_a = 0.1$ and $p_a = 0.2$ — only 68 and 57 out of 201 participants, respectively, report

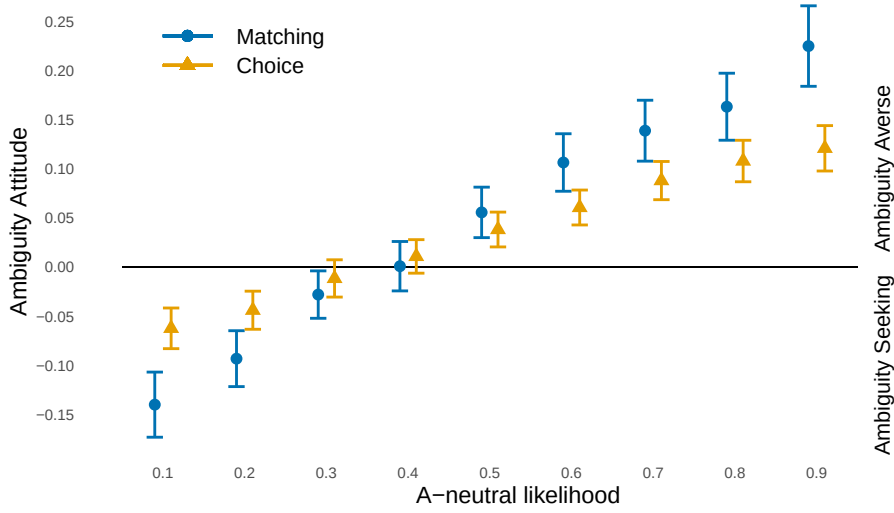


Figure 4: Event-Specific Ambiguity Attitude Indexes by Response Format. Notes: Matching denotes the MP choice list format and Choice denotes the binary choice format. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals. See corresponding statistics in Table B.1.

identical AA_{p_a} under both methods, meaning that over two thirds exhibit inconsistent attitudes. As reported in Table 1, among those who disagree, 62.4% (56.9%) assign a higher AA_{p_a} under MP than under Choice. The pattern reverses for large a-neutral probabilities: at $p_a = 0.8$ and $p_a = 0.9$, the number of consistent participants drops further — to only 44 and 41 out of 201 — with approximately four fifths exhibiting inconsistent attitudes, and 54.1% (58.1%) of them now assigning a higher AA_{p_a} under Choice than under MP. Therefore, the within-subject evidence further reinforces our findings. A large majority of participants report inconsistent AA_{p_a} across binary Choice and MP, especially at the likelihood extremes where a-insensitivity is most pronounced. Moreover, the direction of these individual-level differences follows the predicted inverse-S pattern: relative to binary Choice, MP generates lower matching probabilities for high-likelihood ambiguous events and higher matching probabilities for low-likelihood ambiguous events.

The global indexes lead to the same conclusion. Table 2 reports the summary statistics of the two global indexes across elicitation methods. Both the a-insensitivity index a and the ambiguity aversion index b are positive under both MP and Choice, indicating that participants exhibit likelihood insensitivity and ambiguity aversion on average regardless of how preferences are elicited. Notably, the estimates of a and b under MP (0.43 and 0.09) closely replicate those reported in Dimmock et al. (2016) (0.41 and

Table 1: Instability by event likelihood.

p_a	$AA^{\text{MP}} < AA^{\text{BC}}$	$AA^{\text{MP}} = AA^{\text{BC}}$	$AA^{\text{MP}} > AA^{\text{BC}}$
0.1	50	68	83
0.2	62	57	82
0.8	85	44	72
0.9	93	41	67

Notes: p_a denotes the event’s a-neutral probability. $N = 201$ per p_a level. The table reports the number of participants whose AA derived from Matching is lower than, equal to, or higher than that derived from Choice. Full results across all a-neutral probability levels are reported in Table B.2.

Table 2: Summary statistics of a-insensitivity and ambiguity aversion indices by elicitation method.

Method	A-insensitivity index a			Ambiguity aversion index b			N
	Mean	Median	SE	Mean	Median	SE	
Matching	0.431	0.297	0.032	0.091	0.044	0.018	201
Choice	0.231	0.145	0.020	0.066	0.056	0.013	201

Notes: For a-insensitivity index $a = 1 - s$ and the ambiguity aversion index $b = 1 - s - 2c$, s and c are the slope and intercept of a best-fitting line $mp(\hat{p}) \mapsto c + s\hat{p}$ estimated between each participant’s matching probabilities and the corresponding A-neutral likelihoods. A positive a indicates a tendency to compress likelihoods toward 50-50; a positive b indicates ambiguity aversion on average. SE denotes the standard error of the mean.

0.12). Turning to the cross-method comparison, the ambiguity aversion index b does not differ significantly across the two methods (0.091 *vs.* 0.066; $p > 0.1$), suggesting that the average elevation of the matching function is robust to the elicitation format. By contrast, the a-insensitivity index a is substantially larger under MP than under Choice (0.43 *vs.* 0.23, $p < 0.01$), echoing the stronger a-insensitivity pattern we document in the event-specific analysis.¹⁰ Finally, the cross-method Spearman correlations reveal a moderate agreement for the ambiguity aversion index b ($\rho = 0.49$) but a substantially lower agreement for the a-insensitivity index a ($\rho = 0.20$). It confirms that the main elicitation effect is not a uniform shift in ambiguity aversion, but rather a change in likelihood sensitivity, exactly as implied by the noisy-coding account, with the DM’s prior beliefs unaffected by the stimulus presentation format.

Result 1. *Both binary Choice and MP exhibit ambiguity-insensitivity, and the format of choice list amplifies this pattern.*

¹⁰The results are robust to our random-effect regression after controlling for several personal variables (see Table B.3). Interestingly, the regression results reveal that the a-insensitivity index a negatively correlates with individual cognitive ability, while the ambiguity aversion index b is higher among females.

While a-insensitivity in standalone Choice is already predicted by the Bayesian inference model through asymmetric confidence weights — with ambiguous signals internally processed with greater imprecision than risky probabilities, whose likelihoods are fully transparent — the two-stage cognitive procedure induced by the matching probability structure further compounds this distortion by first transforming the noisier ambiguous signal into a compressed internal anchor, against which the matching probability is calibrated in the second stage.

Decision Noise. The low cross-method correlation in responses — particularly in the a-insensitivity index a — raises the question of whether one method produces noisier data than the other. Given that the format of choice list visually organize those binary choices and group them by the ambiguous lotteries, some may argue that the structured list format helps participants organize their choices, thereby yielding cleaner preference measures. In contrast, others would say that standalone Choice can better capture preferences given its simplicity and free from strategic complexity. Our analysis reveals the difference in data quality and confirms that the binary Choice method produces less decision noise, exactly as our model predicts.

We first leverage those choices between a degenerate lottery and a nondegenerate lottery to examine the direct stochastic dominance violation. Specifically, when the ambiguous lottery is degenerate, out of the nine choices, participants choose the dominated risky one for 2.30 times when they answer in a choice list, compared to 1.04 times when they answer in separate pages ($p < 0.001$). Similarly, when the dominant lottery is risky, participants choose the dominated ambiguous one for 1.04 times in MP, compared to the lower figure of 0.38 in Choice ($p < 0.001$).

Our experiment also includes a repeated MP choice list whose ambiguous lottery has an a-neutral likelihood equal to 0.2 in MP and repeat the corresponding 10 binary choices in the Choice task. Overall, participants exhibit significantly more reversals under the MP format than under standalone Choice (1.36 *vs.* 0.79, $p = 0.003$). In addition, we use Fisher’s exact test to compare the switching rate — the proportion of participants whose choices differ between the two repetitions — at each of the ten known probability levels. The two methods exhibit comparable stability at low likelihood levels ($p_a \leq 0.3$), but Choice produces significantly fewer switches than MP

at intermediate to high likelihood levels ($p_a = 0.4$ to 0.9 , all $p < 0.05$), with the gap widening as p_a increases.¹¹

A complementary perspective on data quality comes from examining multiple switching behavior within the 10×10 choice matrix. Each column constitutes a MP choice list, in which the varying option’s attractiveness changes monotonically as the participant moves down the list; a rational participant should therefore switch at most once per column. We refer to violations of this as *within-list* multiple switching. Each row does not form an explicit list, but the same logic applies horizontally: the fixed option is the risky lottery while the ambiguous lottery varies, and a consistent participant should likewise switch at most once per row. We refer to violations of this as *between-list* multiple switching. As expected, within-list multiple switching is more prevalent under Choice than under MP (1.67 *vs.* 1.16, $p < 0.001$), since the explicit list format nudges participants toward a single switch. More revealing is the between-list pattern: multiple switching is significantly more frequent under MP than under Choice (3.50 *vs.* 1.86, $p < 0.001$), appearing to reflect a redistribution of errors. Specifically, when combining within- and between-list errors, MP induces more monotonicity violations overall than Choice (4.67 *vs.* 3.52, $p = 0.021$). Thus, the apparent improvement in within-list consistency under MP does not translate into higher overall data quality; on the contrary, it suggests that the list structure suppresses local inconsistency within-lists at the cost of generating broader distortions across lists.

Across all three data quality metrics — dominance violations, test-retest reliability, and non-monotonic switching — the evidence consistently points in the same direction: the Choice method elicits less stochastic responses than the MP method.¹²

Result 2. *Standalone Choices yield less noisy and more internally consistent responses than their list-format counterparts.*

This suggests that the stronger a-insensitivity pattern documented under MP does not necessarily provide a more accurate revelation of underlying ambiguity prefer-

¹¹This result is robust to Cohen’s κ , which corrects agreement rates for chance. See Table B.4 for full details.

¹²The ME lists yield the same qualitative pattern: ME elicits statistically more stochastic responses than Choice. It provides further evidence of increased decision noises under list-based elicitation, irrespective of the structure of matching, while none of the differences in corresponding data-quality measures is significant between MP and ME.

ences. Instead, this increase is consistent with the propagation of errors induced by the two-stage decision strategy associated with the list structure: a biased and noisy anchor is formed in the initial stage, whose noise and distortion embedded in this anchor are inherited by the subsequent matching stage of all risky probabilities in the list.

4.3 Matching Event versus Matching Probability

The ME condition serves as a between-subject counterpart to the MP condition, with all procedural details kept identical — the same participant pool, incentive structure, comprehension checks, binary choice task, cognitive ability questions, and demographic questionnaire. The only difference lies in the format of the matching task: whereas the MP lists fix the ambiguous winning range and vary the risky winning range across rows, the ME lists fix the risky winning range and vary the ambiguous winning range across rows. While the full problem set is identical across the two conditions, under any standard preference-based theory of ambiguity, this structural equivalence implies that the distribution of elicited ambiguity attitudes should be indistinguishable between the MP and ME conditions: if preferences are fixed primitives of the individual, it should make no difference whether one moves through the matrix row by row or column by column. The ME condition thus provides an unusually clean and direct test of preference-based theories — one that does not rely on between-subject differences in stimuli, incentives, or framing. Moreover, the ME elicitation is itself a methodological contribution: to our knowledge, no prior study has implemented a matching event procedure in this structurally symmetric fashion, and the design opens a new avenue for assessing how robust measured ambiguity attitudes are to the direction of preference elicitation.

To facilitate direct comparison across the two matching elicitation directions (MP *vs.* ME), we define an analogous probability-specific ambiguity attitude index for the ME elicitation. Recall that under MP, the event-specific index is defined as $AA_{p_a}^{\text{MP}} = p_a - m(p_a)$, where p_a is the a-neutral likelihood of the fixed ambiguous option and $m(p_a)$ is the elicited matching probability. Under ME, the roles of the two options are reversed: the risky winning probability p_r is fixed, and the participant's matching event

— the ambiguous winning range at which she is indifferent — is identified from the switch point in the list. We define the corresponding index as

$$AA_{p_r}^{\text{ME}} = m'(p_r) - p_r$$

where $m'(p_r)$ denotes the a-neutral likelihood associated with the elicited matching event. Ambiguity aversion (seeking) again corresponds to a positive (negative) value: an ambiguity-averse participant requires a wider ambiguous winning range to be willing to accept it, implying $m'(p_r) > p_r$ — the ME analog of requiring a lower matching probability under MP. Crucially, because both the event-specific $AA_{p_a}^{\text{MP}}$ and the probability-specific $AA_{p_r}^{\text{ME}}$ are expressed as deviations from the a-neutral benchmark on the same probability scale, the two indexes are directly comparable across conditions, enabling a clean assessment of how the direction of preference elicitation affects measured ambiguity attitudes.

Figure 5 plots the ambiguity attitude indexes $AA_{p_r}^{\text{ME}}$ alongside the corresponding $AA_{p_a}^{\text{MP}}$ indexes for reference. The ME results are strikingly consistent with the theoretical prediction: the pattern is precisely reversed relative to MP. Under ME, $AA_{p_r}^{\text{ME}}$ is positive for small likelihoods — indicating ambiguity aversion for unlikely events — and turns negative for large likelihoods — indicating ambiguity seeking for likely ones. This constitutes the mirror image of the a-insensitivity pattern documented under MP, and we refer to it as *a-oversensitivity*: rather than compressing subjective likelihoods toward fifty-fifty, participants in the ME condition exhibit greater sensitivity to likelihood differences than the a-neutral benchmark prescribes. The reversal is not a borderline effect — it spans the entire likelihood range, and the $AA_{p_r}^{\text{ME}}$ profile crosses zero at $p_r \approx 0.5$, somewhat above the crossing point of $p_a \approx 0.4$ observed under MP, roughly consistent with the symmetric two-stage account developed in Section 3.

So far, a simple “middle-bias” hypothesis appears able to account for our experimental results. Our evidence, however, reveals a richer picture. When the fixed option in the list is degenerate, participants violate stochastic dominance significantly more often if that option is ambiguous than if it is risky.¹³ This indicates that participants are thus better able to recognize the degeneracy of the fixed option when it carries a known probability than when it is ambiguous — consistent with the model’s premise

¹³For the number of mistaken choices, 2.30 *versus* 1.28, $p < 0.01$; ; for the proportion of participants who violated dominance at least once, 0.46 *vs.* 0.25, $p < 0.01$.

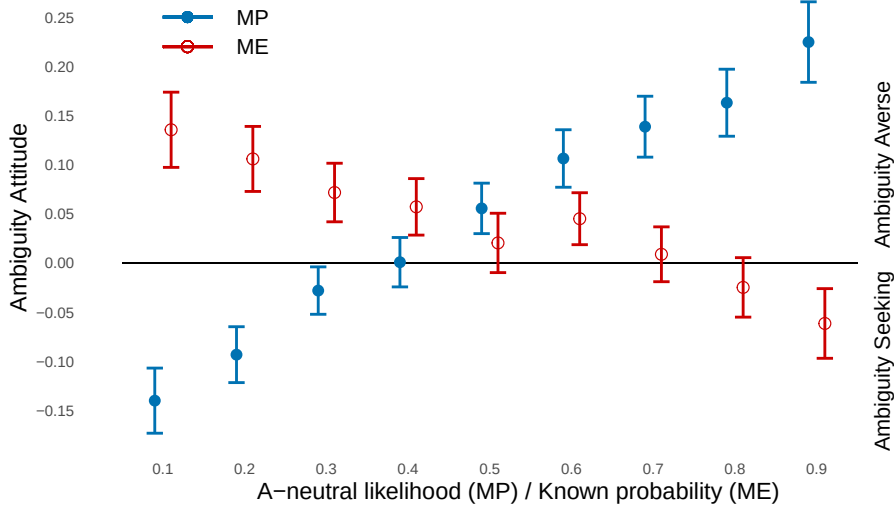


Figure 5: Event-Specific Ambiguity Attitude Indexes by Elicitation Direction. Notes: MP denotes the matching probability choice list and ME denotes the matching event choice list. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals. See corresponding statistics in Table B.5.

that ambiguous probabilities are perceived with greater noise than objective ones.¹⁴

Building on this asymmetry in perceptual precision, our model further predicts that the magnitude of a-oversensitivity under ME should be smaller than that of a-insensitivity under MP — a prediction that yields more nuanced insight and distinguishes our model from such a simple heuristic. Figure 5 is consistent with this prediction: the AA_{pr}^{ME} profile is visibly flatter than its MP counterpart, with the extreme values at the tails lying closer to zero under ME than the corresponding values under MP. To quantify this asymmetry at the individual level, we first examine the individual average ambiguity non-neutrality, defined as the absolute value of the individual mean ambiguity attitude index $|\overline{AA}_i|$. A non-parametric test confirms a significantly lower level under ME than under MP (0.15 *vs.* 0.18, $p = 0.02$), providing a first piece of evidence for the prediction. We further construct a *likelihood sensitivity deviation* index,

$$LSD_i = |AA_i(0.9) - AA_i(0.1)|$$

which captures the spread of a participant’s ambiguity attitude profile across the likelihood range: a larger value indicates a more pronounced deviation from a-neutrality, whether in the direction of a-insensitivity (as under MP) or a-oversensitivity (as under

¹⁴Notably, the two methods differ neither in self-reported overall complexity nor in time spent per list, suggesting that both types of matching list impose a similar cognitive burden overall. The difference in dominance-violation rates is therefore more likely to reflect the difficulty of first-stage evaluation specifically than general task complexity.

ME). The model predicts $LSD^{MP} > LSD^{ME}$. A non-parametric comparison confirms this: the distribution of LSD_i is significantly higher in the MP condition than in the ME condition (0.39 vs. 0.27, $p < 0.01$).¹⁵

Result 3. *ME generates a pattern of ambiguity-oversensitivity, with the distortion directionally reversed and weaker in magnitude relative to MP.*

Under any standard preference-based theory of ambiguity, the MP and ME indexes should be statistically indistinguishable: the two conditions pose identical choice problems, differing only in the direction along which participants traverse the same 10×10 matrix. The systematic sign reversal we observe — a-insensitivity under MP, a-oversensitivity under ME — is therefore direct evidence against such theories. It aligns precisely, however, with the noisy coding account: the direction of the first-stage anchor sets the direction of the distortion, and its weaker magnitude under ME reflects the model’s asymmetric-precision mechanism, whereby MP anchors on the noisier ambiguous signal and ME on the more precisely encoded risky one. The ME results thus do more than replicate the MP–Choice divergence in a new format: they provide a prospective, out-of-sample validation of the two-stage mechanism, testing a prediction no preference-based theory anticipates and that requires the two-stage structure specifically — simpler one-stage noisy coding accounts generate no such reversal.

5 Discussion

This paper develops a noisy coding model of ambiguity attitude elicitation and tests its predictions across three elicitation formats. MP elicits substantially more pronounced a-insensitivity than Choice, with a large share of participants reversing the sign of their measured attitude across the two formats. Across all three data quality metrics, Choice produces cleaner and more internally consistent responses, consistent with the interpretation that the list structure amplifies cognitive distortions rather than reducing them. ME produces a-oversensitivity — the mirror image of the a-insensitivity pattern under MP — with a smaller magnitude of deviation, providing an out-of-sample

¹⁵The result is robust to the alternative specification $LSD_i = |AA_i(0.8) - AA_i(0.2)|$, and to a linear mixed-effects specification that covers all likelihood levels simultaneously, confirming that the magnitude asymmetry is not confined to the tails but persists across the full likelihood range.

validation of the two-stage mechanism. Each of these findings is predicted by the noisy coding model and cannot be accommodated by any preference-based theory of ambiguity.

Implications. Our findings carry three implications. First, the ME sign reversal constitutes the most direct challenge to preference-based accounts of ambiguity: MP and ME present identical decision problems, so any such theory must predict identical measured attitudes. The reversal we document cannot be accommodated by any reparameterization — no adjustment to the probability weighting function or utility curvature can flip the sign of measured attitudes while holding the problem set constant. This implies that a-insensitivity is not a stable preference primitive that elicitation merely reveals, but a pattern whose direction and magnitude are determined by the cognitive architecture of the elicitation format. A further implication is that weaker deviations from neutrality should not be read as evidence of greater rationality: ME elicits smaller deviations than MP not because it is less distorting, but because both impose the same two-stage compression and differ only in the direction it runs.

Second, the noisy coding model endogenizes the central weighting objects of popular descriptive frameworks — such as the capacity function (Schmeidler, 1989) and the source function (Fox and Tversky, 1995; Abdellaoui et al., 2011; Baillon et al., 2025) — as outputs of a Bayesian encoding process. In addition to accounting for source dependence in these parameters, our model explains why they shift with the elicitation format and generates quantitative predictions that our data broadly support. Seen through this lens, cognitive accounts hold the promise of endogenizing the parameters that descriptive ambiguity models treat as primitives, improving our ability to predict when and how measured attitudes respond to payoff-irrelevant features of the elicitation environment (cf. Bouchouicha et al., 2025; Vieider, 2025).

Third, our model implies that a-insensitivity and a-oversensitivity are products of cognitive limitations rather than stable preference primitives. This is consistent with our finding that participants with higher cognitive ability exhibit stronger likelihood sensitivity, and with existing evidence linking cognitive ability to ambiguity attitude measurement (Baillon et al., 2018; Choi et al., 2022). Studies linking ambiguity attitudes to field outcomes such as stock market non-participation or insurance uptake

(Dimmock et al., 2016; Bianchi and Tallon, 2019; Belissa et al., 2020; Anantanasuwong et al., 2024) should therefore reconsider the causal interpretation of those associations: if a-insensitivity is partly a cognitive artifact, documented correlations with field behavior may reflect underlying cognitive ability as much as ambiguity attitudes per se.

Middle bias? One might propose a simpler heuristic account: a “middle bias” whereby participants anchor toward the center rows regardless of the elicitation task. First, our noisy coding model, instead of rejecting its empirical relevance, provides a cognitive micro-foundation for precisely this pull-to-center behavior (e.g., Enke and Graeber, 2023; Enke et al., 2026): compression toward the prior naturally generates list-center anchoring as a special case. Second, a uniform middle bias cannot account for the whole picture. Most critically, it predicts equal compression under MP and ME — our data reject this: the magnitude of deviation from ambiguity neutrality is significantly smaller under ME than under MP, in the precise direction predicted by the encoding precision asymmetry. A heuristic that operates uniformly on list structure cannot generate this quantitative asymmetry; our model does. As another evidence rejecting such simple heuristics, Qiu et al. (2026) show that participants do not blindly switch at the midpoint: over 80% fail a degenerate lottery valuation task yet switch near the correct position rather than the center, suggesting partial directional awareness even when responses are distorted.

Recovering Latent Preferences. Two directions for revealed preference studies follow from our findings. The first concerns elicitation design. A recurring intuition in the literature is that choice lists offer a transparent elicitation format, helping participants organize their decisions by presenting options in a structured sequence (e.g., Tversky and Kahneman, 1992). Our results challenge this view: the list format introduces a two-stage evaluation structure that amplifies cognitive distortions rather than reducing them, making it heavier — not lighter — than the standalone binary choices it subsumes. The key principle is that a more reliable preference elicitation should minimize cognitive burden as much as possible, regardless of how structured it appears. Consistent with this, Chapman et al. (2025) find that a small number of dynamically sequential binary choices elicits preferences more accurately and with greater test-retest stability than standard choice list methods. More broadly, cognitive burden in pref-

erence elicitation is shaped by many design dimensions beyond choice format — including visual presentation, interactive feedback mechanisms, and the sequencing of problems — and systematic attention to each of these factors is a productive direction for future work.

The second direction concerns the recovery of latent preferences from procedure-contaminated data. Several recent approaches offer partial traction: eliciting normative commitments directly (Nielsen and Rehbeck, 2022), leveraging response time to separate deliberate from impulsive choices (Alós-Ferrer et al., 2021), exploiting internal consistency conditions to identify reliable responses (Enke and Graeber, 2023; Bernheim et al., 2026; Enke and Graeber, 2026), and using simple diagnostic tasks to screen response reliability before studying preference patterns (Qiu et al., 2026). A systematic framework for recovering underlying preferences from behavior distorted to an unknown and individually varying extent, however, remains an open methodological challenge to the field.

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Appendices

A A Noisy Coding Model of Ambiguity Attitude

A.1 Baseline Binary Choice Under Ambiguity

We build on the statistical model of ambiguity attitudes developed by [Vieider \(2025\)](#). Consider a DM choosing between a risky lottery that pays a prize x with objective winning probability p_r and an ambiguous lottery that pays the same prize with unknown winning probability. The DM encodes her beliefs about each winning probability as a Beta distribution: $p_a = \text{Be}(\alpha_a, \beta_a)$ for the ambiguous option and $p_r = \text{Be}(\alpha_r, \beta_r)$ for the risky option, where the subscripts a and r stand for ambiguity and risk, respectively.

Following the symmetry of the experimental design across the two options, we assume that the two belief distributions have the same mean: $\hat{p}_a = \hat{p}_r$, where $\hat{p}_a \triangleq \mathbb{E}[p_a] = \frac{\alpha_a}{\alpha_a + \beta_a}$ and $\hat{p}_r \triangleq \mathbb{E}[p_r] = \frac{\alpha_r}{\alpha_r + \beta_r}$.¹⁶ The two belief-formation processes differ only in the source of uncertainty: for the risky lottery, the winning probability is objectively known, whereas for the ambiguous lottery it must be inferred. The key assumption is therefore that beliefs about the ambiguous probability are more dispersed around the true mean than beliefs about the risky probability, reflecting that inferring an unknown event likelihood is cognitively less confident than representing an objective probability. In terms of the concentration parameters of the Beta distributions, this requires $\kappa_a \triangleq \alpha_a + \beta_a < \kappa_r \triangleq \alpha_r + \beta_r$. Equivalently, it implies the variance of the ambiguous distribution is wider, that is, $\mathbb{V}[p_a] > \mathbb{V}[p_r]$, a DM is less certain about the ambiguous probability and encoded with greater imprecision than risky probabilities.¹⁷

The central premise of noisy coding is that DMs cannot directly access probabilities or event likelihoods with perfect precision. Instead, they form noisy internal representations of these quantities and combine them with prior expectations in a Bayesian manner. Following [Vieider \(2024, 2025\)](#), it is computationally convenient to first map

¹⁶The shared-mean assumption is imposed without loss of generality. It is not necessary for the results derived below, but it allows us to isolate the key mechanism generating the model's predictions and is therefore maintained for parsimony.

¹⁷For a given mean, concentration and variance capture the same notion of belief precision because they are monotonically related. In particular, $\mathbb{V}[p_a] \triangleq \frac{p(1-p)}{\kappa_a + 1}$ and $\mathbb{V}[p_r] \triangleq \frac{p(1-p)}{\kappa_r + 1}$.

probabilities into log-odds space:

$$\ell(p) \triangleq \log \frac{p}{1-p} = \log \left(\frac{\alpha}{\beta} \right)$$

Representing probabilities in log-odds space is also consistent with the neural coding paradigm, according to which choice-relevant quantities are encoded in log-odds units (Green and Swets, 1966; Gold and Shadlen, 2001, 2002). Accordingly, $\log \left(\frac{\alpha_a}{\beta_a} \right)$ and $\log \left(\frac{\alpha_r}{\beta_r} \right)$ can be interpreted as unbiased best estimates of the relative odds of winning associated with the ambiguous and risky lotteries, respectively. Then before entering the choice process, each winning odd is internally represented through a noisy signal, denoted by r_a for the ambiguous lottery and r_r for the risky lottery:

$$r_a \sim \mathcal{N}(\ell(p_a), \nu_a^2), \quad r_r \sim \mathcal{N}(\ell(p_r), \nu_r^2)$$

Following the well-known characterization of the logit-normal distribution by Atchison and Shen (1980), the variance of each log-odds signal is determined by the corresponding Beta parameters through the trigamma function $\psi'(\cdot)$:

$$\nu_a^2 \triangleq \psi'(\alpha_a) + \psi'(\beta_a), \quad \nu_r^2 \triangleq \psi'(\alpha_r) + \psi'(\beta_r).$$

Since the ambiguous belief distribution has a lower concentration than the risky belief distribution as we discussed before, this implies $\nu_a^2 > \nu_r^2$, reflecting the greater coding noise attached to the ambiguous signal.

The noisy signal is then combined with a subjective belief about the distribution of log-odds in the environment, given by a common Bayesian prior $\mathcal{N}(\mu, \sigma^2)$ for both risk and ambiguity.¹⁸ This perception process yields the actionable posterior belief that enters the decision stage. Bayesian inference implies that the posterior expectation is a precision-weighted average of the noisy signal and the prior mean:

$$\mathbb{E}[\ell(p_a) | r_a] = \gamma r_a + (1 - \gamma)\mu$$

$$\mathbb{E}[\ell(p_r) | r_r] = \rho r_r + (1 - \rho)\mu$$

where $\rho \triangleq \frac{\sigma^2}{\sigma^2 + \nu_r^2}$ and $\gamma \triangleq \frac{\sigma^2}{\sigma^2 + \nu_a^2}$ denote the posterior weights placed on the perceived signals relative to the prior mean. These weights are determined by signal precision relative to prior uncertainty. Since the ambiguous signal is encoded less precisely than

¹⁸This assumption is consistent with the idea that the average probability—the “best guess”—is the same for the risky and ambiguous urns. Although it is not required for the predictions derived below, imposing a common prior restricts the set of mechanisms through which the observed behavior can arise.

the risky signal, $\nu_a^2 > \nu_r^2$, it follows that

$$0 < \gamma < \rho < 1.$$

Thus, the ambiguous signal is processed with less confidence than the risky signal and is correspondingly shrunk more strongly toward the prior.

In a standalone binary choice between the two options, the DM evaluates the risky and ambiguous lotteries independently and simultaneously. Because the two lotteries offer the same payoff, the DM chooses the risky lottery whenever its posterior expected log-odds exceed those of the ambiguous lottery:

$$\mathbb{E}[\ell(p_r) \mid r_r] > \mathbb{E}[\ell(p_a) \mid r_a]$$

Substituting the posterior expectations derived above, this condition holds if and only if

$$\rho r_r + (1 - \rho)\mu > \gamma r_a + (1 - \gamma)\mu$$

or, equivalently, if and only if the internal representations satisfy

$$\rho r_r - \gamma r_a > (\rho - \gamma)\mu$$

Under our hypothesis about the internal representations, r_r and r_a are independently distributed normal random variables, so that

$$\rho r_r - \gamma r_a \sim \mathcal{N}(\rho\ell(p_r) - \gamma\ell(p_a), \rho^2\nu_r^2 + \gamma^2\nu_a^2)$$

This yields the following stochastic choice rule:

$$\Pr[(x, p_r; 0) \succ (x, E_a; 0)] = \Phi\left(\frac{\rho\ell(p_r) - \gamma\ell(p_a) + (\gamma - \rho)\mu}{\sqrt{\rho^2\nu_r^2 + \gamma^2\nu_a^2}}\right)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The denominator

$$V_{BC} = \rho^2\nu_r^2 + \gamma^2\nu_a^2 \tag{5}$$

captures the variance of corresponding decision noise.

Abstracting momentarily from decision noise, for each ambiguous event E_a with ambiguity-neutral probability p_a , its matching probability $\bar{p}_r$ under the binary choice format can be characterized by the stochastic indifference point. At this point, the probability of choosing the risky lottery is equal to 0.5, which requires

$$\rho\ell(\bar{p}_r) - \gamma\ell(p_a) + (\gamma - \rho)\mu = 0$$

After rearrangement, this condition can be expressed as

$$\ell(\bar{p}_r) = \mu + \frac{\gamma}{\rho} (\ell(p_a) - \mu)$$

Since $0 < \gamma/\rho < 1$, the matching probability inferred from the binary choice is a compressed transformation of the ambiguity-neutral probability around the prior μ : low ambiguous probabilities are shifted upward toward the prior, whereas high ambiguous probabilities are shifted downward toward the prior, thereby generating a -insensitivity endogenously.

A.2 Choice List Elicitation

A.2.1 MP Choice List

In an MP list, the DM faces a sequence of risky lotteries with objective winning probabilities $p_{r_1} < p_{r_2} < \dots < p_{r_J}$, each of which is benchmarked against the same ambiguous lottery. The ambiguous lottery is fixed throughout the list and is associated with an ambiguity-neutral probability p_a . Since the ambiguous option is prominently displayed and remains constant across rows, it is natural to assume that the DM first evaluates this fixed option and then searches for the risky probability that best matches its perception.

Stage-1: Evaluation of the fixed lottery. In the first stage, the DM forms an initial evaluation of the fixed ambiguous lottery. As in the standalone binary-choice environment, this evaluation is based on the posterior inference of its log-odds, given by

$$\ell(p_a) \mid r_a \sim \mathcal{N}(\gamma r_a + (1 - \gamma)\mu, \sigma_{post}^2)$$

where $\sigma_{post}^2 = \frac{\sigma^2 v_a^2}{\sigma^2 + v_a^2}$.

Stage-2: Evaluation of the risky lotteries relative to the anchor. In the second stage, the Stage-1 posterior is carried forward as an anchor-induced prior for the log-odds of each risky lottery. That is,

$$\ell(p_r) \sim \mathcal{N}(\gamma r_a + (1 - \gamma)\mu, \sigma_{post}^2)$$

Given this anchor-induced prior, for each row in the varying list with risky probability p_r , the posterior mean of the risky lottery's log-odds is calculated as

$$\mathbb{E}[\ell(p_r) | r_r] = \hat{\rho}r_r + (1 - \hat{\rho})[\gamma r_a + (1 - \gamma)\mu]$$

where $\hat{\rho} = \frac{\sigma_{post}^2}{\sigma_{post}^2 + \nu_r^2}$. Accordingly, the DMs chooses the risky option whenever $\mathbb{E}[\ell(p_r) | r_r] > \mathbb{E}[\ell(p_a) | r_a]$, which derives:

$$\hat{\rho}r_r > \hat{\rho}[\gamma r_a + (1 - \gamma)\mu]$$

Because $\hat{\rho} > 0$, it can be divided out from both sides. The choice condition therefore reduces to

$$r_r - \gamma r_a > (1 - \gamma)\mu$$

Following the same derivation as in the standalone binary-choice environment, the probability that the DM chooses the risky lottery is therefore

$$\Pr[(x, p_r; 0) \succ (x, E_a; 0)] = \Phi\left(\frac{\ell(p_r) - \gamma\ell(p_a) - (1 - \gamma)\mu}{\sqrt{\nu_r^2 + \gamma^2\nu_a^2}}\right)$$

The variance of the latent decision noise under the MP format is captured by the denominator as

$$V_{MP} = \nu_r^2 + \gamma^2\nu_a^2 \quad (6)$$

Thus, under the same structural normalization used for the standalone binary-choice rule in (5), since $\rho < 1$ always holds, the MP procedure clearly generates greater decision noises than the standalone binary-choice procedure. This increase is consistent with the propagated errors generated from the two-stage decision strategy induced by the list structure: a biased and noisy anchor is formed in the initial stage, whose noise and distortion are then inherited in the subsequent matching stage.

By looking at the indifference point under a continuous approximation, the matching probability in choice list is represented by

$$p_r^* = \arg \min_{p_r} \|r_r | p_r - [\gamma\ell(p_a) + (1 - \gamma)\mu]\|$$

It implies that in the second stage, the DM searches across the risky probabilities in the list for the row whose internal signal most closely matches the ambiguous anchor inferred from Stage 1. Formally, this search problem selects the risky probability p_r^* whose log-odds match the ambiguous anchor on average. This procedure differs from the standalone binary-choice environment. In Choice, the risky and ambiguous op-

tions are evaluated separately and compared directly. In an MP list, by contrast, the fixed ambiguous option is first transformed into an internal anchor, and the DM then searches for the risky probability that produces indifference with that anchor.

The elicited matching probability p_r^* is characterized by

$$\ell(p_r^*) = \mu + \gamma (\ell(p_a) - \mu)$$

Since $0 < \gamma < 1$, this mapping contracts the ambiguity-neutral log-odds toward the prior mean μ , implying that MP responses also exhibit a-insensitivity. Under standalone binary choice, the corresponding contraction coefficient is γ/ρ . Since $0 < \rho < 1$, it follows that $\gamma < \gamma/\rho$. Therefore, the contraction toward the prior mean is stronger under MP than under standalone binary choice. Consequently, MP lists amplify a-insensitivity relative to standalone binary choice.

A.2.2 ME Choice List

In an ME list, the matching direction is reversed. The fixed option is now a risky lottery with objective winning probability p_r , whereas the varying options are ambiguous lotteries associated with ambiguity-neutral probabilities $p_{a_1} < p_{a_2} < \dots < p_{a_j}$.

As in the MP list, the fixed option is evaluated first and subsequently serves as an internal anchor for the search across the varying rows. The crucial distinction is that, under ME, the fixed option is risky rather than ambiguous. The DM therefore searches across the ambiguous events and selects the event whose ambiguity-neutral probability p_a^* most closely matches the internal representation of the fixed risky lottery.

In Stage-1, the fixed risky lottery is evaluated according to the posterior distribution

$$\ell(p_r) | r_r \sim \mathcal{N}(\rho r_r + (1 - \rho)\mu, \sigma_{post}^2), \quad \sigma_{post}^2 = \frac{\sigma^2 \nu_r^2}{\sigma^2 + \nu_r^2}$$

and serves as the anchor-induced prior in Stage-2 for the varying ambiguous probabilities. Then DM selects the ambiguous row whose internal signal most closely matches this risky anchor. Accordingly, the latent ME response is centered at

$$p_a^* = \arg \min_{p_a} \|r_a | p_a - [\rho \ell(p_r) + (1 - \rho)\mu]\|$$

The latent matching event is therefore characterized by

$$\ell(p_r) = \mu + \frac{1}{\rho} (\ell(p_a^*) - \mu)$$

Since $1/\rho > 1$, the ME-list induces oversensitive ambiguity attitudes—a reversed pattern relative to both the binary choice and the MP-list—ambiguity aversion for unlikely events and ambiguity seeking for likely events. Additionally, the insensitivity generated by the MP list is governed by $1 - \gamma$, whereas the oversensitivity generated by the ME list is governed by $1 - \rho$. Since $0 < \gamma < \rho < 1$, the model predicts stronger prior-centered compression under MP than under ME. This asymmetry arises because MP anchors on the ambiguous fixed option in stage 1, whose signal is more heavily shrunk toward the prior, whereas ME anchors on the risky fixed option, whose signal is less shrunk.

The stochastic choice rule under ME is given by

$$\Pr[(x, p_r; 0) \succ (x, E_a; 0)] = \Phi \left(\frac{\rho \ell(p_r) - \ell(p_a) + (1 - \rho)\mu}{\sqrt{\rho^2 \nu_r^2 + \nu_a^2}} \right)$$

The variance of the latent decision noise under the ME format is therefore

$$V_{ME} = \rho^2 \nu_r^2 + \nu_a^2 \quad (7)$$

Thus, as in the MP procedure characterized by Equation (6), the ME procedure also generates greater latent decision noise than standalone binary choice.

B Additional Figures and Tables

Table B.1: Event-Specific Ambiguity Attitude Indexes by Elicitation Method

	A-neutral likelihood $\hat{p}$								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MP	−0.140	−0.093	−0.028	0.001	0.056	0.106	0.139	0.163	0.225
Choice	−0.062	−0.044	−0.012	0.011	0.038	0.060	0.088	0.108	0.121
<i>p</i> -value	<0.001	<0.001	0.197	0.399	0.189	0.004	0.002	0.003	<0.001

Notes: Each cell reports the mean $AA_{\hat{p}} = \hat{p} - mp(\hat{p})$ for the indicated elicitation method and a-neutral likelihood $\hat{p}$. Positive (negative) values indicate ambiguity aversion (seeking). *p*-values are from two-sided paired *t*-tests of the null hypothesis that $AA_{\hat{p}}$ is equal across MP and Choice.

Table B.2: Response frequency by event likelihood and response category.

	A-neutral probability $\hat{p}$								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$AA^{\text{Matching}} < AA^{\text{Choice}}$	50	62	59	50	63	78	84	85	93
$AA^{\text{Matching}} = AA^{\text{Choice}}$	68	57	63	74	75	65	59	44	41
$AA^{\text{Matching}} > AA^{\text{Choice}}$	83	82	79	77	63	58	58	72	67

Notes: $\hat{p}$ denotes the event's A-neutral probability. AA^{Matching} represents the *AA* derived from Matching responses, while AA^{Choice} represents the *AA* derived from Choice responses. $N = 201$ per $\hat{p}$ level.

Table B.3: Regression results for a-insensitivity and ambiguity aversion indices.

	A-insensitivity (<i>a</i>)		Ambiguity aversion (<i>b</i>)	
	(1)	(2)	(3)	(4)
Intercept	0.23*** (0.03)	0.29*** (0.10)	0.07*** (0.02)	−0.05 (0.07)
Matching	0.20*** (0.03)	0.20*** (0.03)	0.03 (0.02)	0.03 (0.02)
Order (DESC)		0.02 (0.04)		0.01 (0.03)
Age		0.00 (0.00)		0.00 (0.00)
Female		0.06 (0.04)		0.05* (0.03)
Cognition		−0.01* (0.01)		0.00 (0.00)
$\Delta_{Complexity}$		−0.00 (0.02)		−0.02 (0.02)

Notes: All models are estimated with random intercepts for participants. The variable Matching is a dummy equal to 1 for observations elicited via the Matching method. $\Delta_{Complexity}$ denotes the difference in perceived complexity between the two elicitation formats. Standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B.4: Test-retest reliability of elicitation methods by known probability level.

$\hat{p}$	Switching Frequency		Fisher p	Cohen's κ	
	Matching	Choice		Matching	Choice
0.1	33	24	0.253	0.580	0.452
0.2	48	43	0.634	0.515	0.573
0.3	38	35	0.796	0.503	0.439
0.4	37	13	0.000	0.406	0.571
0.5	29	14	0.023	0.372	0.463
0.6	18	7	0.037	0.558	0.517
0.7	16	6	0.046	0.556	0.557
0.8	19	6	0.012	0.461	0.556
0.9	19	3	0.001	0.405	0.762

Notes: $N = 201$ per $\hat{p}$ level. Switching rate denotes the number of participants whose choice reverses between the two repetitions. Fisher p denotes the p -value from Fisher's exact test comparing switching rates across methods. Cohen's κ measures agreement between the two repetitions within each method, corrected for chance.

Table B.5: Mean Ambiguity Attitude Index across Likelihood Levels by Elicitation Task

	A-neutral likelihood (MP) / Known probability (ME)								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MP	-0.140	-0.093	-0.028	0.001	0.055	0.106	0.139	0.163	0.225
ME	0.135	0.106	0.072	0.057	0.020	0.045	0.009	-0.025	-0.062

Notes: For the MP task, each cell reports the mean $AA_p^{MP} = \hat{p} - mp(\hat{p})$, where e denotes the A-neutral likelihood of the ambiguous lottery and $mp(e)$ is the elicited matching probability of the risky lottery. For the ME task, each cell reports the mean $AA_p^{ME} = \hat{p}_{me(p)} - p$, where p denotes the known probability of the risky lottery and $mp(p)$ is the elicited matching probability of the ambiguous lottery. Positive (negative) values indicate ambiguity aversion (seeking).

C Robust Analysis

When the analysis is restricted to the 344 of 400 participants who passed all three comprehension questions correctly on their first attempt, the main findings are replicated, as shown in Figures C.1 and C.2.

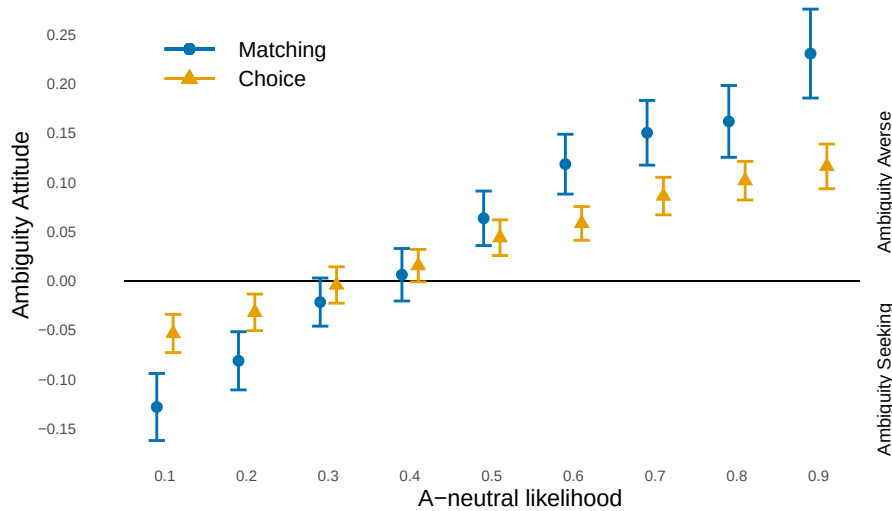


Figure C.1: Event-Specific Ambiguity Attitude Indexes by Response Format. Notes: Matching denotes the MP choice list format and Choice denotes the binary choice format. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals.

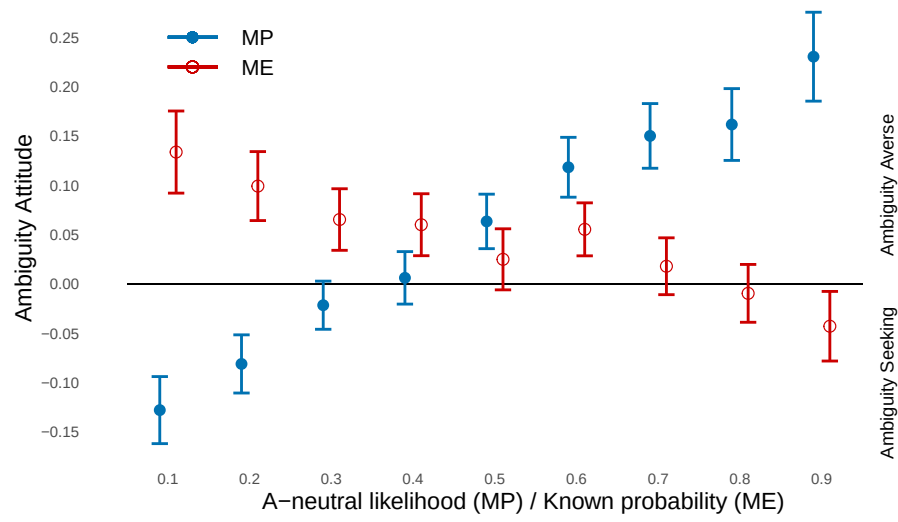


Figure C.2: Event-Specific Ambiguity Attitude Indexes by Elicitation Direction. Notes: MP denotes the matching probability choice list and ME denotes the matching event choice list. Positive (negative) values indicate ambiguity aversion (seeking). Error bars denote 95% confidence intervals.